Sparse Disapproval Guarantees a Nonempty Hare Core

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Abstract

Whether every approval-based committee election admits a deterministic committee in the Hare core remains open. We prove nonemptiness on the full co-rank-two domain: every voter may disapprove at most two candidates, while the numbers of candidates, seats, and voter types are otherwise unrestricted. The proof gives an explicit selection rule. Among all missing sets of the required size, first maximize the weight of two-disapproval voters whose disapproval pair is hit, and then maximize total disapproval incidence. A random-set inequality rules out targets one seat below the committee, while the secondary tie-break rules out unanimous equal-size targets. Smaller targets cannot improve any voter. Two independent exact implementations enumerate every target for more than two hundred thousand maximizing committees on complementary finite grids. These computations audit, but are not needed for, the unbounded symbolic theorem.

1 Introduction

An approval election consists of a finite voter set $\mathcal{N}$, a candidate set C of size m, approval sets $A_i \subseteq C$, and a target committee size k. A nonempty set $T \subseteq C$ blocks a size-k committee W at the Hare quota when

$$\sum_{i:\,|A_i \cap T| > |A_i \cap W|} w_i \geq \frac{|T|}{k}, \tag{1}$$

where the positive rational voter weights sum to one. A committee is in the Hare core if it has no blocker.

The core was introduced as a strong proportional-representation requirement for approval committees by Aziz et al. [1]. General deterministic nonemptiness is open. The current parameter frontiers include $k \leq 8$ and $m \leq 15$ [2], and at most seven weighted voter types [3]. Other positive results impose interval or related preference structure [5]. Berker et al. [4] give an exact mixed-integer/duality framework and additional special cases.

Our restriction is orthogonal to these bounds. Write

$$H_i = C \setminus A_i$$

for voter i's disapproval set. We assume only $|H_i| \leq 2$. Thus the number of voter types may grow without bound, and the result applies in the open parameter range, for example at $(m, k) = (17, 9)$ when every voter approves at least fifteen candidates. A targeted primary-source audit using "bounded disapproval," "two-veto," "dense approvals," and "complement rank" terminology found no equivalent theorem; the novelty conclusion is necessarily bounded and is not a claim of absolute priority.

2 Selection rule and main theorem

The endpoint $m = k$ is immediate by taking $W = C$. Assume henceforth that $m > k \geq 2$ and put $s = m - k \geq 1$. A committee is represented by its missing set $B = C \setminus W$, where $|B| = s$. Define

$$e(B) = \sum_{i:\,|H_i|=2} w_i \, \mathbf{1}[H_i \cap B \neq \varnothing], \qquad j(B) = \sum_i w_i |H_i \cap B|. \tag{2}$$

Choose a size-s set B lexicographically maximizing $(e(B), j(B))$ and return $W = C \setminus B$.

Theorem 1 (Co-rank-two core theorem)**.** *For every $m \geq k \geq 2$, every approval election in which $|C \setminus A_i| \leq 2$ for every positive-weight voter has a nonempty Hare core. When $m > k$, every committee selected by rule (2) is in the core.*

The rule is explicit and finite. We do not claim a polynomial-time implementation; exhaustive evaluation over all $\binom{m}{s}$ missing sets is sufficient for the mathematical statement and the supplied verifier.

3 Proof

Let T be a nonempty target of size $t \leq k$. Write $T = C \setminus D$ and $d = k - t$. Then $|D| = s + d$. Direct cancellation gives, for every voter,

$$|A_i \cap T| - |A_i \cap W| = -d + |H_i \cap D| - |H_i \cap B|. \tag{3}$$

3.1 Targets of size at most $k - 2$

If $d \geq 2$, the right side of (3) is at most $-d + |H_i| \leq 0$. No voter strictly gains, so such a target cannot block.

3.2 Targets of size $k - 1$

Now $d = 1$. Strict gain in (3) requires

$$|H_i \cap D| - |H_i \cap B| \geq 2.$$

Thus every gainer has $|H_i| = 2$, has $H_i \subseteq D$, and has $H_i \cap B = \varnothing$. In particular, every gainer is a two-disapproval voter not counted by $e(B)$.

Lemma 2. *A uniformly random size-s set meets any fixed two-set with probability*

$$p_{m,s} = 1 - \frac{k(k-1)}{m(m-1)} > \frac{1}{k}. \tag{4}$$

Proof. The avoidance probability is $\binom{m-2}{s}/\binom{m}{s} = k(k-1)/(m(m-1))$. With $m = k + s$,

$$k[m(m-1) - k(k-1)] - m(m-1) = (k-1)((2s-1)k + s(s-1)) > 0,$$

which is exactly the strict inequality in (4). □

Let w_2 be the total weight of two-disapproval voters. Averaging (2) over random B' and using maximality gives $e(B) \geq p_{m,s} w_2$. Hence every size-$(k-1)$ target has gainer weight at most

$$w_2 - e(B) \leq (1 - p_{m,s}) w_2 < 1 - \frac{1}{k} = \frac{k-1}{k}.$$

This is strictly below its Hare quota.

3.3 Targets of size k

Here $d = 0$, so blocking requires every positive-weight voter to gain. Suppose that happened for some other size-s missing set D. Equation (3) yields

$$|H_i \cap D| > |H_i \cap B| \qquad \text{for every } i. \tag{5}$$

An empty H_i makes (5) impossible. For each two-disapproval voter, (5) implies that every pair hit by B is also hit by D, and every pair missed by B is hit by D. Therefore $e(D) \geq e(B)$. Strict inequality contradicts the first lexicographic objective. If equality holds, summing (5) over all voters gives $j(D) > j(B)$, contradicting the tie-break. Thus no size-k target blocks. Together with the two preceding cases this proves Theorem 1.

4 Exact verification

The primary verifier represents disapproval types by immutable sets and anonymous count vectors. On its sealed grid it checks 101,851 profile–parameter cases, 236,801 tied lexicographic maximizers, and 12,020,946 literal committee–target inequalities. The independent verifier uses bitmasks and a separate enumeration architecture; it checks 106,620 cases, 265,371 maximizers, and 15,499,790 literal inequalities, including the boundary $(m, k) = (3, 2)$. Both construct every tied maximizer, not merely one.

All arithmetic is integral after clearing rational voter denominators. The finite grids are regression evidence. Lemma 2 and the proof above establish the unbounded quantifiers.

5 Scope and open boundary

The theorem permits arbitrary positive rational weights, repeated ballots, unapproved candidates, and targets overlapping the committee. Equality in the Hare quota counts as blocking throughout. The theorem does not cover a voter disapproving three candidates, does not resolve co-rank three, and does not imply unrestricted core nonemptiness. The natural next question is whether a higher-order coverage objective can control deficit-one targets without losing the equal-size tie-break; the present averaging proof alone does not supply such an extension.

Reproducibility and AI assistance

The package README gives a one-command exact replay and PDF build. Generative AI systems were used during proof exploration, verifier development, literature-query formulation, and manuscript preparation. The author reviewed the statements, proofs, citations, and executable artifacts and takes responsibility for the content.

References

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